

Student Performance Prediction

Mini Project Report

Predict student academic performance using machine learning

Project Domain	Machine Learning / Educational Analytics
Application	Student Performance Prediction Web Interface
Input Factors	Attendance, Study Hours, Assignment Score, Previous Exam Marks, Sleep Hours

Prepared as a mini project documentation report

1. Project Objective

The Student Performance Prediction project is designed to estimate a student's academic performance from a small set of measurable academic and lifestyle factors. The application provides a simple web-based form where student details are entered and then used as inputs for a machine-learning prediction workflow.

The interface shown in the supplied screenshots focuses on collecting the required inputs in a clear, student-friendly format. The selected inputs are student name, attendance percentage, study hours per day, assignment score, previous exam marks, and sleep hours.

2. Key Features

- Clean web interface for entering student information.
- Student name field for identifying the prediction record.
- Attendance percentage input.
- Study hours per day input.
- Assignment score input.
- Previous exam marks input.
- Sleep hours input.
- Next-step workflow for moving from data entry to prediction.
- Prediction output can be connected to a trained machine-learning model using the collected features.

3. Tools & Technologies

Component	Technology / Concept
Programming	Python
Machine Learning	Supervised machine-learning model for performance prediction
User Interface	Web-based form interface
Input Processing	Numeric feature collection and validation
Project Environment	Local development environment

4. Working / Methodology

Step 1 — Enter student details

The user enters the student's name and the numerical values requested by the interface.

Step 2 — Validate inputs

The application checks that the required fields contain usable values before continuing.

Step 3 — Prepare model features

The numerical inputs are arranged in the same feature order used when the prediction model was trained.

Step 4 — Generate prediction

The trained machine-learning model receives the feature vector and produces the predicted academic-performance result.

Step 5 — Display output

The application presents the prediction in the output stage so that the user can interpret the student's expected performance.

5. Input Parameters

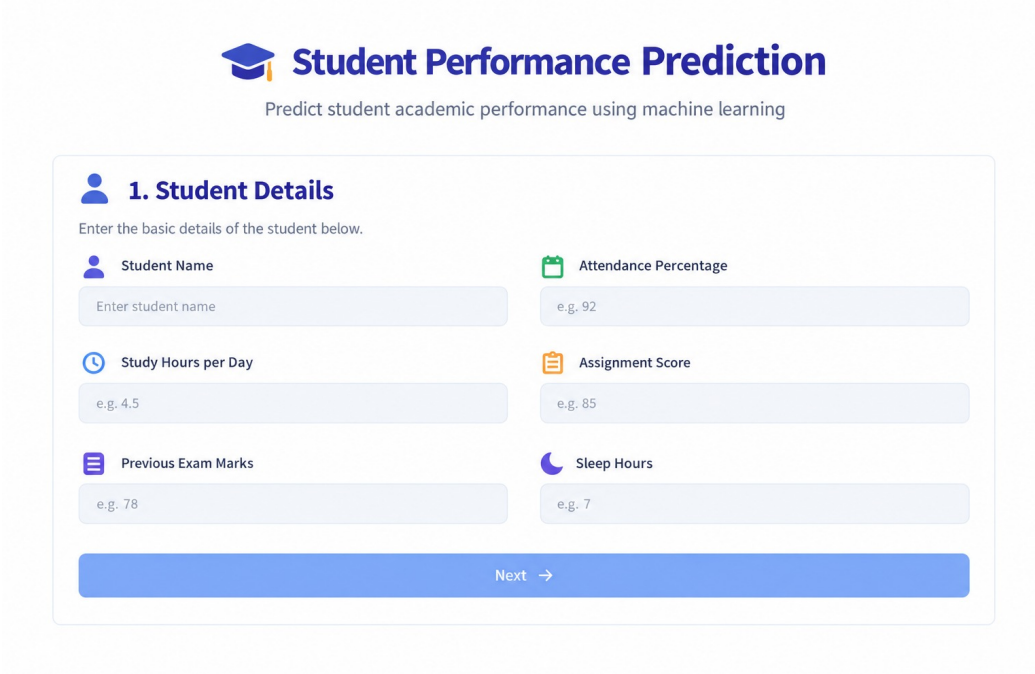
Input	Purpose
Student Name	Identifies the student record.
Attendance Percentage	Represents the student's attendance level.
Study Hours per Day	Represents the average daily study time.
Assignment Score	Represents assignment performance.
Previous Exam Marks	Provides a prior academic-performance indicator.
Sleep Hours	Represents the student's average sleep duration.

Note: The uploaded reference material describes a different project (AI Interview Preparation System). For this report, the content has been adapted to the supplied Student Performance Prediction interface and its visible fields; a specific trained-model algorithm, dataset, accuracy, or exact prediction formula was not provided in the supplied material.

6. Output / Application Screenshots

The following screenshots are included as the visual output/reference for the Student Performance Prediction mini project. The first image shows the empty input form, while the second shows the same form populated with sample student values.

Fig. 1 — Student Performance Prediction: Input Form



The screenshot displays a web application titled "Student Performance Prediction" with the subtitle "Predict student academic performance using machine learning". The main section is titled "1. Student Details" and includes the instruction "Enter the basic details of the student below." The form consists of six input fields arranged in two columns: "Student Name" (placeholder: "Enter student name"), "Attendance Percentage" (placeholder: "e.g. 92"), "Study Hours per Day" (placeholder: "e.g. 4.5"), "Assignment Score" (placeholder: "e.g. 85"), "Previous Exam Marks" (placeholder: "e.g. 78"), and "Sleep Hours" (placeholder: "e.g. 7"). Each field is preceded by a small icon representing its category. At the bottom of the form is a blue button labeled "Next →".

The screen contains fields for Student Name, Attendance Percentage, Study Hours per Day, Assignment Score, Previous Exam Marks, and Sleep Hours, followed by a Next button.

Fig. 2 — Student Performance Prediction: Sample Input

The screenshot displays a mobile application interface titled "Student Performance Prediction" with the subtitle "Predict student academic performance using machine learning". The interface is divided into sections. The first section, "1. Student Details", prompts the user to "Enter the basic details of the student below." It contains six input fields arranged in a 3x2 grid, each with an icon and a label: "Student Name" (person icon) with the value "Aliya", "Attendance Percentage" (calendar icon) with the value "75", "Study Hours per Day" (clock icon) with the value "2", "Assignment Score" (clipboard icon) with the value "80", "Previous Exam Marks" (document icon) with the value "80", and "Sleep Hours" (moon icon) with the value "8". At the bottom of the form is a large blue button labeled "Next →".

Field	Value
Student Name	Aliya
Attendance Percentage	75
Study Hours per Day	2
Assignment Score	80
Previous Exam Marks	80
Sleep Hours	8

Sample values visible in the supplied screenshot: Student Name — Aliya; Attendance — 75%; Study Hours — 2 per day; Assignment Score — 80; Previous Exam Marks — 80; Sleep Hours — 8.

7. Expected Prediction Output

After the user clicks Next, the application can pass the entered values to the trained model and display the predicted academic-performance result. A complete model output screenshot was not supplied, so this report does not invent a predicted score or performance category for the sample student.

8. Advantages

- Simple and easy-to-use student data entry interface.
- Uses multiple academic and lifestyle indicators rather than a single input.
- Can provide a quick prediction once connected to a trained model.
- Can be extended with charts, performance categories, and recommendation messages.

9. Future Enhancements

- Train and compare multiple machine-learning algorithms.
- Add model accuracy, confusion matrix, and evaluation metrics to the project report.
- Show a prediction category such as expected performance level if defined by the project model.
- Add personalized study recommendations based on the input values.
- Store previous predictions for student progress tracking.
- Add data visualizations for attendance, study time, marks, and predicted performance.

10. Conclusion

The Student Performance Prediction mini project demonstrates how a web interface can collect relevant student information and use those features as inputs to a machine-learning prediction workflow. The supplied screenshots provide a clear representation of the input stage and sample data entry. With a trained model connected to the workflow, the application can be extended into a complete end-to-end academic performance prediction system.